

Cyrus Guided Notes: Stanford CS231A Computer Vision, From 3D Reconstruction to Recognition

Stanford course pack

2026-05-15

Course source

Official source: <https://web.stanford.edu/class/cs231a/>

Why this course is in the system. Entry point for camera models, projection, epipolar geometry, optical flow, estimation, and 3D reconstruction.

Learning output. Already connected to local course notes and a guided PDF.

Prerequisite checkpoint

- Linear algebra
- Projective geometry
- Probability and estimation

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\begin{aligned} s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} &= K[R \ \mathbf{t}] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \\ \mathbf{x}'^T F \mathbf{x} &= 0 \\ E &= [\mathbf{t}]_{\times} R \\ \min_{T_i, P_j} \sum_{i,j} \rho(\|\mathbf{u}_{ij} - \pi(T_i P_j)\|_2^2) \end{aligned}$$

Visual page

Show a 3D point, camera pose, image plane, and projection ray.

GoodNotes pages

- Page ST-CV001: camera projection
- Page ST-CV002: epipolar geometry
- Page ST-CV003: bundle adjustment

Web interaction

A draggable projection lab showing how depth changes image coordinates.

Obsidian and Notion

Layer	Output
Obsidian	Stanford Spatial AI -> CS231A
Notion	Geometry notes and PDF evidence.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.